KPSS Test

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Introduction

The Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test is the natural complement to the augmented Dickey-Fuller (ADF) test for assessing stationarity. The two tests have flipped null hypotheses: ADF tests a unit-root null against a stationary alternative; KPSS tests a stationary null against a unit-root alternative. Running both and comparing their conclusions gives a more robust verdict than either alone, because a single test that fails to reject is uninformative — failure to reject a unit-root null is not evidence of stationarity, and vice versa.

Prerequisites

A working understanding of stationarity, unit-root processes (random walks), and the augmented Dickey-Fuller test as the complementary stationarity diagnostic.

Theory

The KPSS framework decomposes $y_t = r_t + \beta t + \varepsilon_t$, where $r_t = r_{t-1} + u_t$ is a random walk. Under the null of stationarity, the variance of the random-walk innovations is zero ($\sigma_u^2 = 0$); under the alternative, r_t contributes a stochastic trend. The test statistic is based on partial sums of residuals from regressing y_t on (optionally) a constant and a trend; large partial-sum behaviour indicates a unit root and the test rejects stationarity.

The "Level" version tests stationarity around a constant mean; the "Trend" version tests stationarity around a deterministic trend. Choosing between them depends on whether the series has a visible time trend.

Assumptions

Correct model specification (level vs trend stationarity); the lag truncation parameter is adequate for the underlying autocorrelation structure (default works for most applications).

R Implementation

```r
library(tseries)

# Stationary series
y1 <- rnorm(100)
kpss.test(y1, null = "Level")

# Random walk (unit root)
y2 <- cumsum(rnorm(100))
kpss.test(y2, null = "Level")

# Trend-stationary series
y3 <- 0.5 * (1:100) + rnorm(100)
kpss.test(y3, null = "Trend")
```

Output & Results

`kpss.test()` returns the KPSS statistic and the p-value (with a note about the truncation lag and the type of stationarity tested). Small statistics correspond to large p-values and indicate insufficient evidence to reject stationarity; large statistics reject and support a unit root.

Interpretation

A reporting sentence: "The stationary white-noise series gave a KPSS statistic of 0.08 ($p > 0.10$, fail to reject stationarity); the random-walk series gave a statistic of 2.34 ($p < 0.01$, reject stationarity)." Always state which null was tested (Level vs Trend) and pair with an ADF test for the complementary perspective.

Practical Tips

- `null = "Level"` for plain stationarity around a constant mean; `"Trend"` for stationarity around a deterministic trend. Choose based on whether a time-trend is visible in the series.
- Bandwidth (lag-truncation) choice affects test size; the default is adequate for most series. For long-memory or strongly autocorrelated series, increase the truncation manually.
- Combine with the ADF test: if both agree (ADF rejects unit root, KPSS fails to reject stationarity), stationarity is well-supported; if both disagree (ADF fails, KPSS rejects), a unit root is well-supported.
- If both tests give ambiguous answers (ADF fails, KPSS fails), the series is borderline and additional diagnostics (autocorrelation structure, Phillips-Perron, fractional integration) help.
- For multivariate or panel data, use Im-Pesaran-Shin or Levin-Lin-Chu tests; KPSS is univariate.
- Differencing the series and re-testing is the standard workflow if a unit root is detected; the differenced series should pass both KPSS and ADF.

R Packages Used

`tseries::kpss.test()` for the canonical implementation; `urca::ur.kpss()` for an alternative with finer control over deterministic components and bandwidth selection; `tseries::adf.test()` for the complementary ADF test; `forecast::ndiffs()` for an integrated workflow that automatically applies KPSS to determine the required differencing order.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics